

Probabilistic dengue and chikungunya forecasting in Brazil

Team Neural Earth — Methodology, IMDC 2026

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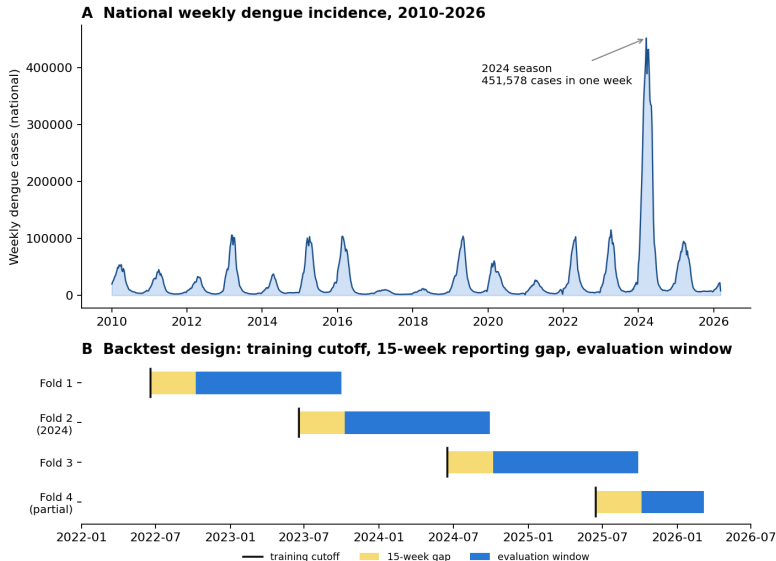
Internal methodology webinar, 22 September 2026

The problem

- Brazil's 2024 dengue season broke the national record by roughly $4\times$: 451,578 cases in a single epidemiological week, $\sim 6.67\text{M}$ for the year.
- Preparedness needs **calibrated probabilistic forecasts**, not point estimates: the full range of plausible outcomes, not just an expected value.
- Open question this challenge exposes: which modeling paradigm (statistical, ML, DL, mechanistic) forecasts best, and how robustly, when a season can depart by an order of magnitude from the training record?

Our approach: one leakage-safe backtesting harness, five model families, evaluated identically across four official seasons and 26 mandatory states, combined into a conformally-recalibrated ensemble.

Study design



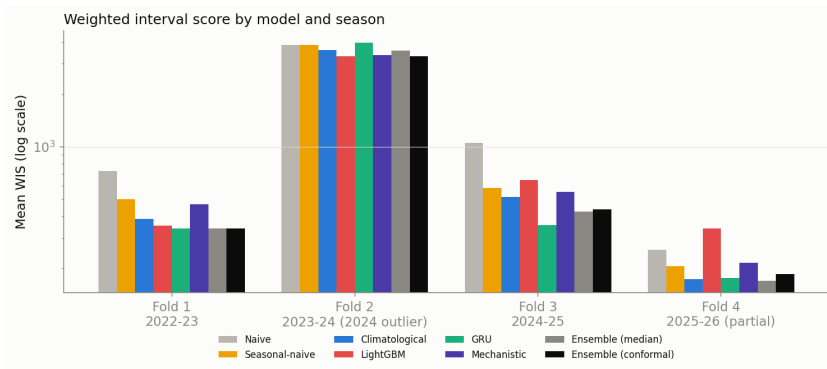
Leakage-safe backtest discipline

- Fold boundaries derived at runtime from the challenge's own `train_N/target_N` flags — never hardcoded, so the pipeline stays correct if the data is refreshed.
- A confirmed **15-week reporting gap** sits between each fold's training cutoff and its target window in every official fold. Every joined covariate table (climate, ocean indices, search-access) is explicitly date-filtered to the cutoff, not inferred from the target flag, so this gap cannot leak into training.
- The ECMWF seasonal-forecast product is filtered by *issue date*, not target date: a legitimate future covariate only if it was issued on or before the cutoff.
- A hard leakage assertion guards the whole pipeline, backed by dedicated regression tests.
- Same machinery, unchanged, produces the real forecast-phase submission (train cutoff = EW25 2026, target EW41 2026 → EW40 2027).

Five model families, one harness

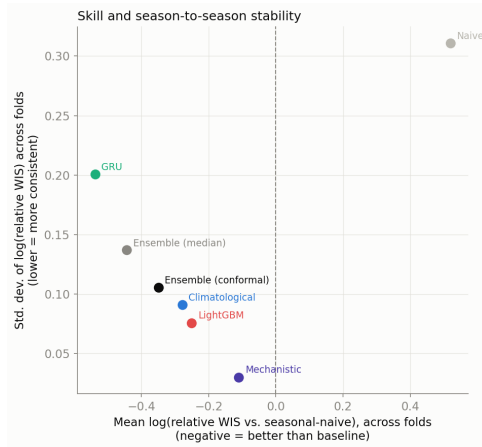
- **Baselines** — naive, seasonal-naive, climatological quantile (empirical same-epiweek quantiles).
- **LightGBM** — quantile regression on a synthetic-origin panel (weekly origins $\times$ horizons 1–67), direct multi-horizon, with conformalized quantile regression (CQR) calibration.
- **GRU** — global recurrent network, per-state embeddings, negative-binomial head, 5-member deep ensemble.
- **Mechanistic** — Richards-curve bootstrap of historical trajectories with a negative-binomial observation model; intervals widen with historical variability by construction.
- **Ensemble** — per-quantile median (Vincentization) of climatological + LightGBM + GRU, then conformal recalibration of the tails.

Finding 1: no single model dominates



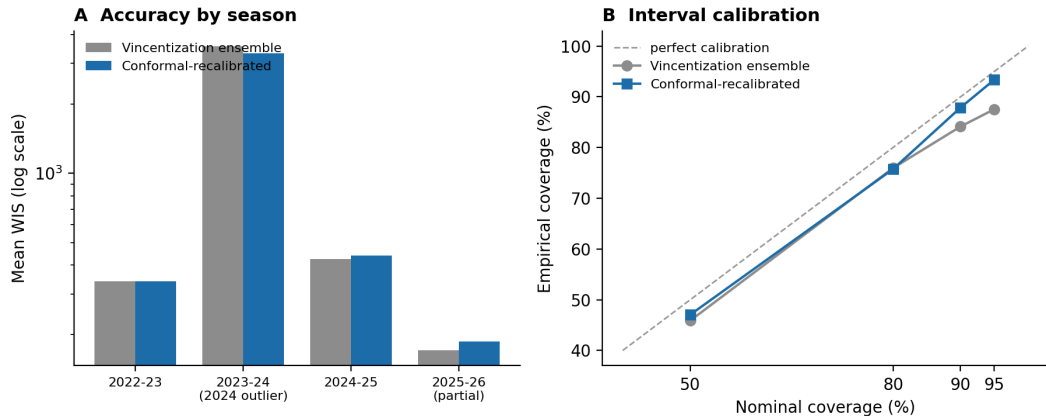
The GRU is best in ordinary seasons, worst on the 2024 outlier. LightGBM shows the opposite profile. The ensemble never suffers a member's worst-season failure.

Skill and stability are different axes



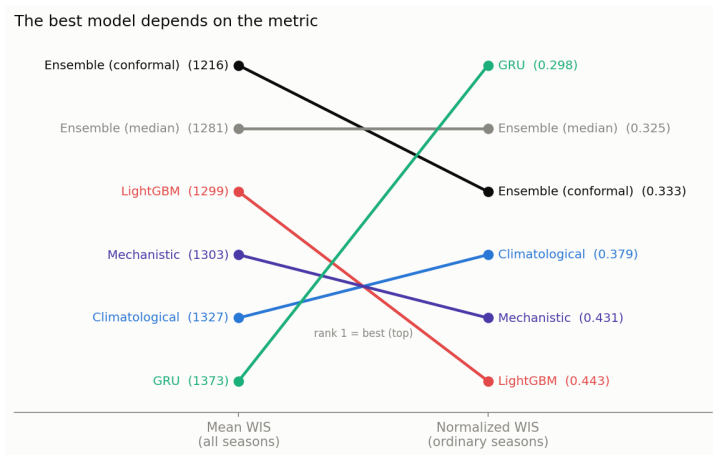
Mean vs. std-dev of log relative WIS across folds. The GRU has the best mean skill but is among the least stable; the mechanistic model is the most stable but modest; the ensembles occupy a position no single member reaches.

Finding 2: conformal recalibration is cheap, transferable insurance



Factors {50: 1.03, 80: 1.00, 90: 1.23, 95: 1.73} tuned on one season, applied to all others. Mean WIS 1281 → 1216 (out-of-sample −5.5%); coverage 46/76/84/88 → 47/76/88/93. Gain concentrates in the outbreak season at negligible cost elsewhere.

Finding 3: the winner depends on the metric

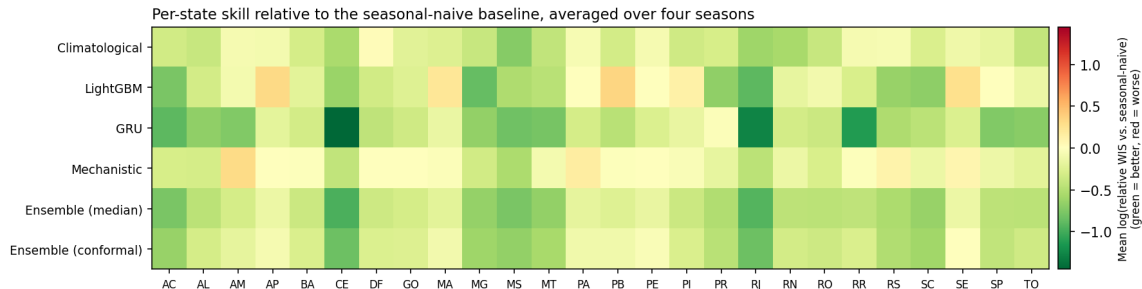


Magnitude-weighted mean WIS and scale-free normalized WIS reward accuracy on different subsets of a highly skewed set of states. We report both; a single aggregate can mislead.

Finding 4: added complexity did not pay

- **Observed climate covariates** added to LightGBM: WIS **worsened** (1299 $\rightarrow$ 1342). The forecast-relevant signal is the ECMWF seasonal forecast and ENSO, not observed weather — observed climate is already implicit in recent case lags.
- **Hierarchical macroregion pooling** of seasonal climatology: WIS worsened (1216 $\rightarrow$ 1241). States are too heterogeneous within a macroregion for naive pooling to help.
- An internal-holdout hyperparameter search that *improved* the holdout *worsened* the true backtest — a caution about tuning discipline in heavily-autocorrelated synthetic-origin panels.
- Simple methods (climatological quantile, a historical-trajectory bootstrap) are hard to beat, consistent with small-sample, few-series forecasting-competition evidence.

Beyond the aggregate: per-state and disease-specific results



Per-state skill relative to the seasonal-naive baseline. The GRU's advantage concentrates in specific states rather than being uniform. Chikungunya: gradient boosting alone beats the ensemble (WIS 82.3 vs. 91.6) — the best model is disease-specific.

Independent recognition and current status

- At the 31 July 2026 validation-round webinar, our submission was named:
 - ▶ one of **6 best-overall-performing models** (of 27) for the mandatory dengue state-level target;
 - ▶ one of **3 models with outstanding performance** on the optional city-level target.
- **Forecast phase (2026–27 season) submitted** 6 September 2026, ahead of the 10 September deadline, using the same harness and the conformal-recalibrated ensemble.
- Public repository: leakage tests, scoring-correctness tests, determinism guards, provenance manifest, and validated submissions for all four official tracks.

Takeaways

- ① Model rankings are season-dependent; a model chosen on average performance can be the least reliable exactly when an outbreak makes accuracy matter most.
- ② An unweighted ensemble is never the single worst model in any season, and conformal recalibration turns it into the single best, at negligible cost.
- ③ Metric choice is a reporting-standards issue, not a modeling one: report both magnitude-weighted and scale-free skill.
- ④ Complexity did not pay here: the signal that helps is the *right* covariate (seasonal forecast, not observed weather), not more of it.

`github.com/kamrul28890/3rd_imdc_purdue_neuralearth`